Distance-local receptor evidence in `vdjmatch`

Algorithm appendix — shipped default, opt-in routes, bounds and observations

Revised 2026-10-11

Abstract

`vdjmatch` annotates T-cell receptors with peptide–MHC specificity by searching a labelled reference (VDJdb) and calibrating every count against a matched background. This appendix is the reviewable account of the algorithm as it stands. Part I gives the intuition, the stage-by-stage flow and a map to the implementation. Part II specifies the shipped default. Part III specifies the opt-in distance-local rank model, its null bound, its Bayesian reading and — newly — the regime in which it degenerates. Part IV covers reference-defined germline rules. Part V states the parameters and why each takes its value. Parts VI and VII derive time, space and accuracy bounds and compare them with recorded observations. Part VIII lists what remains unproved. Exact guarantees, conditional results and heuristics are kept apart throughout. The original scorer remains the default; no experimental route below is a promoted replacement, and no quantity here is a calibrated biological probability of a true specificity.

I. Intuition and flow

1 The problem, in one paragraph

A T-cell receptor recognises a peptide–MHC complex (pMHC) mainly through its CDR3 loops, and receptors recognising the same pMHC converge onto similar loops. So: given a query receptor, find its near neighbours among receptors of *known* specificity, and decide whether that neighbourhood is more crowded than chance would produce. The second half is the whole difficulty. Immune repertoires are enormously redundant and some junctions are generically "popular"; a raw neighbour count therefore says as much about how ordinary a sequence is as about what it binds. Every statistic below is a different answer to *crowded relative to what*.

Inputs. A query q: a TRA receptor, a TRB receptor, or a genuine linked TRA–TRB pair, each carrying an anchor-inclusive junction (the Cys104…Phe118 string, *not* the anchor-free IMGT CDR3) and, where available, V and J gene calls. A reference set of VDJdb source records, each a (junction, V, J, peptide, MHC, study) observation. A background control population: either generated draws from a recombination model (OLGA) or a real repertoire.

Outputs. Per (query, pMHC target) a ranked candidate row carrying an evidence score, the supporting counts, a control-calibrated enrichment p-value, and the population sizes that produced them; per query a top-1 call with an explicit status (`matched`, `ambiguous`, `no_hit`, `no_reference`, `invalid_query`). Distance, database confidence, rank evidence, E-value, conditional model probability and empirical error rate are separate columns and are never silently fused.

2 Stages

1. **Normalise and validate.** Resolve locus, check the junction alphabet, resolve V alleles against the bundled templates, apply reference eligibility and any requested exclusions. Query identity, locus, V/J and pairing survive every join.
2. **Index.** Build one `seqtree` trie per locus over the *distinct* reference junctions. Observation multiplicity is retained separately; it must not multiply a search key.
3. **Search.** One native batch per locus over the distinct query junctions, within a fixed edit budget θ. The same batch runs against the control index, giving the cumulative control neighbour count $n_c(s)$ at every attained score s.
4. **Score.** Reduce hits to one row per (query, pMHC), weighting each neighbour by its similarity and dividing by the local background crowding.
5. **Calibrate.** Compare the reference neighbour count with the count the background predicts, as a Poisson tail.

6. **Call.** Rank candidates; the top one is the call, exact ties abstain.

Steps 3–5 have opt-in alternatives (Part III) that replace the edit ball with a TCRdist geometry and the density with a rank statistic. They are *not* the default.

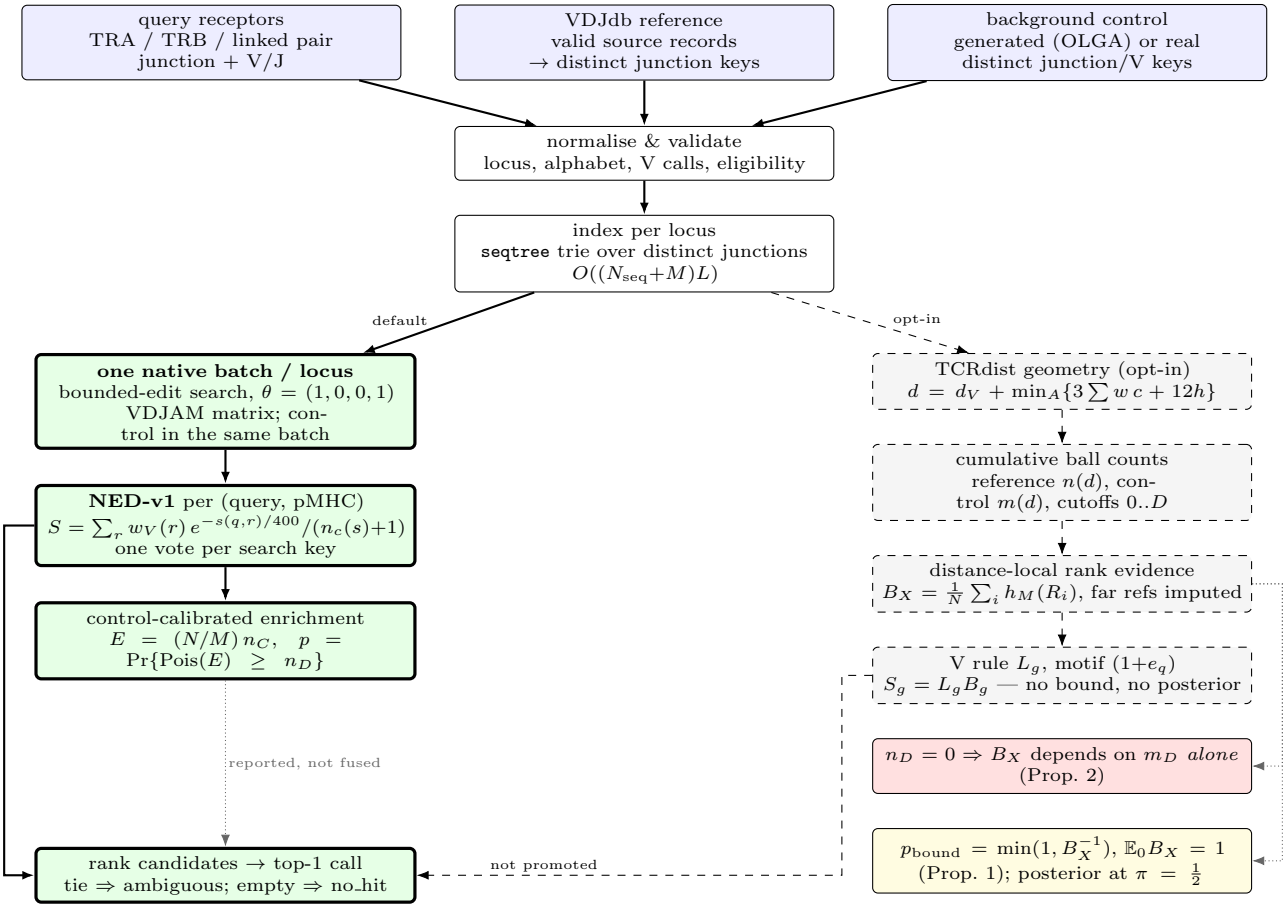


Figure 1: Algorithm flow. **Solid, bold** is the shipped default route; **dashed** is opt-in and not promoted; **dotted** is evidence reported alongside the ranking and never fused into it. Topology source: `figures/algorithm_flow.dot`.

3 Map to the implementation

Stage	Module	Entry point
normalise / validate	`io.columns`, `api`	`normalize_query`, `_prepare`
index + search	`match.engine`	`VdjdbIndex.build`, `.annotate`
scope parsing	`match.scope`	`parse_scope`, `search_params`
substitution matrix	`match.scoring`	`load_vdjam`
positional weights	`match.regions`	`significance_weights`, `position_weights`
NED-v1 reduction	`aggregate.candidates`	`candidates`
enrichment	`evalue.single`, `seqtree.evalue`	`evalue_result`
first-hit (adaptive)	`evalue.first_hit`	`scan`, `pvalue`
paired evidence	`evalue.paired`, `api`	`paired_candidates`
TCRdist geometry	`match.tcrdist`	`distance_matrix`, `total_distance_count_batch`
rank evidence	`evalue.local_rank`	`local_rank_evidence`, `nearest_conditioned_rank`
V rules, motifs	`evalue.epitope_rules`	`reference_epitope_rules`

II. The shipped default

4 Receptors, targets and populations

The query q is a TRA, TRB or linked TRA–TRB receptor. Target X includes peptide and MHC restriction. Its eligible reference set has size N; the declared background has size M. A junction includes both conserved anchors. Full-junction identity exclusion and metadata eligibility precede every count. References and real controls use distinct junction/V keys; generated controls retain draws. Those different sampling units do not establish a common null distribution.

5 Search and the NED-v1 density

Let $\theta = (s_{\max}, i_{\max}, d_{\max}, t_{\max})$ be the edit budget (default $(1, 0, 0, 1)$) and $B_\theta(q) = \{x : \mathrm{edit}(q, x) \preceq \theta\}$ the ball it defines. For each reference neighbour $r \in B_\theta(q)$ the engine returns an alignment score $s(q, r)$ under the bundled VDJAM substitution matrix (integer scale 100). Let $n_c(s)$ be the number of *control* junctions within alignment score s of q — the cumulative control count at that score, obtained from the same batched search. With $w_V(r) \in \{1\} \cup [0, 0.25]$ the soft V-gene weight (Sec. 12), the default per-target evidence is

$$S_{\mathrm{NED}}(q, X) = \sum_{r \in B_\theta(q) \cap X} \frac{w_V(r)\, e^{-s(q,r)/\lambda}}{n_c(s(q, r)) + 1}, \qquad \lambda = 400.$$

The sum runs over distinct search keys (junction, V, J), so publication multiplicity never multiplies a vote; record, clonotype, study and cell counts are reported separately. Reading it: the numerator is similarity, decaying exponentially in alignment cost; the denominator is how crowded the background is at that same distance, so a neighbour reached only in a region where the background is also dense is discounted. The $+1$ is a pseudocount making the statistic finite and size-invariant for a private neighbour, and it is what makes an exact match *counted rather than automatically accepted*.

Implementation: `aggregate.candidates.candidates`. Its `background_mass=True` variant replaces $w_V e^{-s/\lambda}$ by one vote per distinct junction and rescales by M/N; the historical variant of `match.unified` instead divides by $\max\{N\widehat{F}_q^{\mathrm{edit}}(H), 0.01\}$. All three exist; **the shipped default is the boxed form above**.

6 Control-calibrated enrichment

Let n_D be the number of distinct reference neighbours and n_C the number of distinct control neighbours in the same ball. Under an exchangeable background the expected number of reference neighbours is $E = (N/M)\, n_C$ and

$$p_{\mathrm{enrich}} = \Pr\{\mathrm{Poisson}(E) \geq n_D\}.$$

The Poisson/Chen–Stein justification, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix. The quantity is significant only when a receptor has *more* labelled neighbours than the generative background predicts — the signature of antigen-driven convergence rather than of a merely common sequence. It is reported alongside the ranking and never multiplied into it.

First-hit (adaptive) scope. Rather than fix the radius, `evalue.first_hit` widens to each query's *nearest* reference hit, up to five total edits with at most two insertions and two deletions, and evaluates E at that first-hit radius r^*: $E = (N/M) n_C(r^*)$, $p = \Pr\{\mathrm{Poisson}(E) \geq n_D(r^*)\}$. Because the background count grows faster with radius than the target count (measured in Sec. 12), the calibration adapts per query with no hand-tuned scope.

Linked pairs. A paired candidate requires both chains to match the *same* reference complex. Under the background law α/β pairing is close to unconstrained, so the joint null factorises, $E_{\alpha\beta} = N\pi_0^\alpha \pi_0^\beta$, while antigen-driven selection makes the two chains dependent *conditional on an epitope* — which is exactly the excess the joint test detects. Exact-pair puncturing removes only the exact/exact corner. The finite-control bound uses four simultaneous one-sided Clopper–Pearson category bounds with total failure budget $\delta = 10^{-6}$ (`evalue.paired._joint_punctured_results`); δ is added to the reported tail, so the result is a bound, not an exact tail. Deduplicated controls do not establish the IID assumption the bound conditions on.

III. Distance-local rank evidence (opt-in)

7 One additive distance

For residues a, b, let $s(a, b)$ be BLOSUM62 and set $c(a, a) = 0$, $c(a, b) = \max(0, \min(4, 4 - s(a, b)))$ for $a \neq b$. Let h be the junction-length difference, L the longer length, and A an allowed alignment with one central block of h gaps. The TCRdist-like distance is

$$d(q, r) = d_V(q, r) + \min_A \Big\{ 3 \sum_{(i,j)\in A} w_{L,\max(i,j)}\, c(q_i, r_j) + 12h \Big\}.$$

Here d_V sums aligned V-template CDR1/CDR2/CDR2.5 distances, each with weight one; i, j are zero-based positions in the full junctions. TCRdist3 uses unit interior weights, trimming three N-terminal and two C-terminal residues. The symmetric comparison trims three at both ends. Alternatively the shipped empirical profile uses $u_{L,k} = 1 - \max\{p_V(k), p_J(L - 1 - k)\}$ and $w_{L,k} = u_{L,k}/(L^{-1} \sum_t u_{L,t})$. The profiles p_V, p_J are measured same-epitope retention probabilities indexed from each end, not exact NDN boundaries. Native weights have 0.01 resolution/minimum; they weight substitutions.

The term $12h$ is a *linear* gap penalty, not an affine opening/extension scheme or a Smith–Waterman alignment. An affine alternative would be $g(0) = 0$, $g(h) = g_{\text{open}} + (h - 1)g_{\text{extend}}$ for $h > 0$; it is a different model. For linked receptors sum the two chain distances to the *same* reference pair. Do not infer a joint null from independent marginal control chains.

8 Historical score and the broad-kernel experiment

Original vdjmatch sums, over nonexact equal-length neighbours within five substitutions,

$$S_{\text{old}} = \sum_r \frac{a(q, r) e^{-P(q,r)/400}}{\max\{N \widehat{F}_q^{\text{edit}}(H(q, r)), 0.01\}}.$$

Here a is the historical V factor, P the positional BLOSUM Gram penalty, H the substitution count, and $\widehat{F}_q^{\text{edit}}$ the control edit-ball fraction. Sparse targets additionally use the original marginal V/J/length log-frequency ratios; historical paired output uses cohort rank fusion. These contracts remain reproducible and are distinct from the model below.

The preceding experimental score was $S_{\text{linear}} = (W_X/N)/[(W_B + 1/2)/(M + 1)]$, where $W_X = \sum_r K_D(d(q, r))$, $W_B = \sum_b K_D(d(q, b))$ and $K_D(t) = \max(0, 1 - t/D)$ at fixed radius D (not diameter). It compares broad average proximity. A very close match may still be diluted. The construction below restores distance-local normalisation; neither experiment is an accepted replacement default or an established biological TP probability.

Background identity matters. Generative draws measure recombination accessibility; real controls include biological selection; VDJdb excluding X reflects other annotated specificities and ascertainment. Keep these questions separate. Point probability $P_{\text{gen}}(q)$ is not neighbourhood probability. Preserve full junctions for identity/Pgen even if terminal matching weights are zero.

9 A normalised reciprocal-rank score

Rank and ties. For reference r_i, let R_i be its distance rank among the M controls plus itself, so $R_i \in \{1, \ldots, M+1\}$. Let $H_k = \sum_{j=1}^{k} 1/j$ and $H_0 = 0$. Define

$$\boxed{B_X(q) = \frac{1}{N} \sum_{i=1}^{N} h_M(R_i), \qquad h_M(r) = \frac{M + 1}{H_{M+1}\, r}.}$$

This is inverse local background mass with its finite harmonic normalisation, not an exponential kernel with a fitted scale. In the CLI both populations use integer bins $\lceil d_{\text{native}}/200 \rceil$. For l controls in lower bins and t controls in the same bin, average over the possible tied ranks:

$$\bar{h}_{l,t} = \frac{M + 1}{H_{M+1}} \cdot \frac{H_{l+t+1} - H_l}{t + 1}.$$

Do not break ties using target labels or input order.

Censoring far evidence. The fixed radius D determines computation, not a chosen target-dependent gate. Let n_D, m_D be the reference/control counts inside it. Far references contribute their conditional null mean

$$\bar{h}_{\text{far}} = \frac{M+1}{H_{M+1}} \cdot \frac{H_{M+1} - H_{m_D}}{M - m_D + 1}, \qquad B_X = \frac{\sum_{i:d_i \le D} \bar{h}_i + (N - n_D)\bar{h}_{\text{far}}}{N}.$$

No far ordering enters this statistic. No near evidence on either side gives $B_X = 1$. A singleton target is valid. Empty populations are explicitly unavailable; the model posterior then remains its prior. No clustering minimum, K cap or 20-residue switch is required.

Remark 1 (numerical form)**.** $H_{M+1} = \psi(M+2) + \gamma$ and $\bar{h}$ over an inclusive integer interval $[\ell, u]$ is $(\psi(u+1) - \psi(\ell))/(u - \ell + 1)$; `local_rank._harmonic_interval` evaluates widths ≤ 8 by exact summation instead, because the digamma difference cancels catastrophically for large, closely spaced ranks. The explicit assignment $B_X = 1$ when $n_D = m_D = 0$ is algebraically redundant — the general formula already yields 1 — and is a float-safety guard.

10 Null bound and Bayesian derivation

Proposition 1 (expectation-one null)**.** *Under H_0 — distance labels exchangeable conditional on the pooled distances and fixed population sizes — a randomised reference rank is uniform on $\{1, \dots, M+1\}$, so $\mathbb{E}_0 h_M(R_i) = \frac{M+1}{H_{M+1}} \cdot \frac{1}{M+1} \sum_{r=1}^{M+1} \frac{1}{r} = 1$ and hence $\mathbb{E}_0 B_X = 1$. Shared controls require no independence between contributions; tie averaging and genuinely censored far ranks preserve the equality by conditional expectation. Consequently*

$$\boxed{p_{\text{bound}} = \min(1, B_X^{-1}), \qquad \Pr_0(p_{\text{bound}} \le \alpha) \le \alpha \quad (0 < \alpha < 1).}$$

This Markov bound is conservative, not the exact tail distribution of B_X. A separate nearest-reference test has $p_{\text{nearest}} = \Pr\{K \ge 1\}$ for $K \sim \text{Hypergeom}(N + M, N, m_* + 1)$, where m_* counts controls at or below the nearest reference bin. Upper-bin ties are conservative; censored no-hits receive one. It is not combined with the full-score bound.

For the *specified rank-tilted alternative* H_1, define $dP_1 = B_X \, dP_0$. Its mass is one by Proposition 1, so B_X is its Bayes factor. With declared prior $\pi = \Pr(H_1)$,

$$\boxed{\Pr(H_1 \mid \text{rank evidence}) = \frac{\pi B_X}{1 - \pi + \pi B_X}.}$$

The CLI default $\pi = 1/2$ means equal model odds, not biological specificity prevalence. This alternative is explicitly chosen; its normalisation is not proof that it describes receptor recognition. Deduplication, generated-draw multiplicity and ascertainment may violate exchangeability. Outputs therefore remain conditional model quantities until sampling and biological calibration are validated. P-values are not probability FP, and model posteriors are not automatically probability TP. Target/route selection requires its own calibration. General expectation-one evidence: Vovk and Wang, Ann. Statist. 49:1736–1754 (2021), doi:10.1214/20-AOS2020.

11 The unsupported-query degeneracy

The censoring rule is unbiased (Proposition 1) but it is not *informative*, and that distinction decides the statistic's behaviour on sparse targets.

Proposition 2 (degeneracy at zero support)**.** *If $n_D = 0$ — no reference of target X lies within the radius of q — then*

$$B_X(q) = \frac{(M+1)\,(H_{M+1} - H_{m_D})}{(M - m_D + 1)\,H_{M+1}},$$

a strictly decreasing function of the control count m_D alone*. It does not depend on N, on any reference distance, or on anything else specific to X.*

Proof. With $n_D = 0$ the near sum is empty and all N references take $\bar{h}_{\text{far}}$, so $B_X = \bar{h}_{\text{far}}$, which is the displayed expression. Monotonicity in m_D follows because $(H_{M+1} - H_m)/(M - m + 1)$ is the mean of $1/r$ over $[m+1, M+1]$, a mean over a shrinking upper interval. □

Remark 2 (both propositions checked numerically)**.** Against the shipped `local_rank_evidence` itself, not a reimplementation. Proposition 1: over 4,000 exchangeable draws at $N = 40$, $M = 5{,}000$ and 30 shared distance bins, $\widehat{\mathbb{E}_0 B_X} = 1.00385 \pm 0.00828$ (s.e.), i.e. 0.47 standard errors from one. Proposition 2: with $n_D = 0$ the code reproduces the closed form to a relative 10^{-12} over five values of m_D spanning 0 to $M - 1$, and is bit-identical across four N at fixed m_D: the claimed independence of N is exact, not approximate.

Remark 3 (three blocks, and what each is worth)**.** Remark 7 separates queries by whether multi-reference evidence exists. For a linked pair there is a third state, and it is the common one: no *common* reference pair, but per-chain neighbours present. It holds 77.6% of all linked-pair queries and 67.3% of those on the priority targets, and 97.4% of such queries do have per-chain evidence. Ordering that block by the sum of the two chains' within-block ranks of the existing single-chain evidence — a scale-free fusion needing no weight — rather than by the degenerate paired density gives, with the single-chain route left exactly as shipped: pooled mean average precision over 625 tasks 0.8205 → 0.8285 and median 0.8410 → 0.8476 (76 wins, 23 losses); over the 70 priority-target tasks 0.8201 → 0.8297; over all 99 linked-pair tasks 0.8175 → 0.8678; over the 14 priority linked-pair tasks 0.8261 → 0.8738. Using one chain only recovers +0.0241 of the +0.0504 linked-pair gain, so both chains are needed. This is the only construction tested that improves the pooled statistic and the per-target objective simultaneously. It is not promoted; it is recorded with its measurement, and the fresh curated run that would decide it has not been authorised.

Remark 4 (the statistic itself is exhausted)**.** Three reformulations of the rank statistic were tested and none moves the metric. Dropping the far-imputation term from the ranking is mathematically well motivated — a synthetic sweep of the far background moves B_X by 0.215 while the near mass is invariant to 1.4×10^{-14} — yet it changes pooled average precision by +0.0001. Averaging over supported references instead of all of them is worse by 0.0316 pooled. Tie-breaking is vacuous: B_X already resolves to 1.00 distinct values per query at $N = 30$. What moves the metric is how queries are *routed* by the evidence they possess, not how the evidence is summarised once possessed.

Remark 5 (what this costs)**.** Within a one-versus-all task, every query with $n_D = 0$ is therefore ranked purely by how crowded its *background* neighbourhood is — a measure of how generic the receptor is, carrying no information about the label. The damage scales with the fraction of such queries. Measured on the recorded all-task complement run, the median supported fraction is 0.817 for single chains but only 0.173 for linked pairs, because the paired route applies the same numeric radius to the *sum* $d_\alpha + d_\beta$, which is a roughly twofold tighter per-chain constraint. The consequences are quantified in Sec. 12. This is a property of the censoring imputation, not a defect of the rank construction, and it is an open design question (Part VIII), not a fixed parameter choice.

Remark 6 (the paired ball is right, and tightness is why)**.** Linked-pair support requires a *common* reference pair inside the radius: $d_\alpha + d_\beta \le D$ to the same pair. The natural loosening — asking only that each chain independently have some reference neighbour, the criterion the factorised paired E-value uses — is worthless as a support test. At the same $D = 90$, measured over the fourteen priority linked-pair tasks, the common-pair criterion covers 0.593 of positives against 0.054 of negatives (a 11.1× purity ratio) while the independent criterion covers 0.974 against 0.954 (1.0×); over all ninety-nine linked-pair tasks, 32.8× against 1.2×. The tightness of the summed constraint is therefore not an accident of implementation — it is the entire source of paired discriminating power, and the low coverage it causes is better read as a shortage of linked reference pairs than as a radius that needs widening.

Remark 7 (the repair is the disposition, not the statistic)**.** Two corrections are easily confused. Removing unsupported *queries* from a comparison is not the same operation as removing censored *references* from the statistic, and only the first is beneficial. Measured over the recorded all-task run: leaving B_X unchanged and merely ranking $n_D = 0$ queries below every supported query — ordering them among themselves by the existing density — raises mean average precision from 0.8179 to 0.8460 over all 99 linked-pair tasks (69 wins, 23 losses, 7 ties) and from 0.8272 to 0.8776 over the 14 priority linked-pair tasks (12/2). Recomputing B_X over supported references only, with the same block order, instead *loses* 0.0316 mean average precision pooled and 0.0694 on priority single chains, because it discards censored far-reference mass that is genuinely informative once most references are far. The ablation attributes the gain to the block order, not to conditioning the statistic. Neither variant is promoted; both are recorded so the distinction is not re-made.

12 Nearest-conditioned rank

`local_rank.nearest_conditioned_rank` refines exactly the $n_D = 0$ branch. Let k be the number of controls at or below the nearest reference distance. The nearest reference then has control rank exactly $k + 1$; the remaining $N - 1$ references have ranks in $[k + 1, M + 1]$, and the refinement assigns each the mean reciprocal rank over that

interval:

$$B_X^{\text{cond}} = \frac{M+1}{N\,H_{M+1}} \left[\frac{1}{k+1} + (N-1) \cdot \frac{H_{M+1} - H_k}{M-k+1} \right].$$

References with radius support are untouched. Counting all control ties at the nearest distance is conservative.

Remark 8 (proof obligation)**.** Conditional on the observed nearest-reference rank the remaining $N-1$ ranks are *order-constrained*, not uniform on $[k+1, M+1]$. The imputation is therefore a conditional-mean heuristic, and $\mathbb{E}_0 B_X^{\text{cond}} = 1$ is **not** established for this branch. Until it is, Proposition 1 and the bound p_{bound} should not be claimed for queries routed through it. The exposure is not marginal: in the recorded all-task run 99,670 of 614,281 scored queries (16.2%) have $n_D = 0$; that is only 3.0% of single-chain queries but 83.6% of linked-pair queries — precisely where the paired evidence is read. The refinement is a ranking improvement with an unproved null, which is a different object from the supported-query statistic.

IV. Reference-supported V rules and exhaustive CDR3 rank

This opt-in ranking diagnostic keeps germline usage separate from sequence geometry. Let g be the query V gene, or its linked alpha/beta V-gene tuple. Let N, M be the target and other-peptide population sizes, and n_g, m_g their counts in category g, after benchmark-only full-junction puncturing. Populations retain distinct full-junction/resolved-V-allele keys. Let K be the number of observed population V categories plus one unseen category. Define

$$p_g = \frac{n_g + m_g + 1/2}{N + M + K/2}, \qquad L_g = \begin{cases} \dfrac{(n_g + p_g)/(N+1)}{(m_g + p_g)/(M+1)}, & n_g m_g > 0, \\ 1, & n_g m_g = 0. \end{cases}$$

Missing category support on either side contributes no V evidence. For supported categories, calculate the multi-reference statistic B_X using only same-category target/control CDR3 distances and populations n_g, m_g; otherwise use CDR3 distances across all categories with populations N, M. Both routes have zero V-loop cost and trim three full-junction residues at each end. Linked distances compare both query chains with the same reference pair. No reference is censored: distances are multiples of three and are bounded by twelve times the sum of maximum full-junction lengths over the chain axes. The final distance-bin counts must equal the punctured population counts. Write this exhaustive rank statistic as B_g; the ranking score is $S_g = L_g B_g$.

For a motif family, let T_q contain distinct query tokens; linked queries use token tuples from their two chains. Let a_t, b_t count same-category target/control receptors containing token t, with paired presence required within the same linked receptor. For a supported category and nonempty T_q,

$$e_q = \frac{1}{|T_q|} \sum_{t \in T_q} \max\left(0, \frac{a_t}{n_g} - \frac{b_t}{m_g}\right), \qquad S_{g,T} = S_g(1 + e_q), \quad 0 \le e_q \le 1;$$

otherwise $e_q = 0$. Exact three-residue, IMGT-class three-residue, internal-wildcard four-residue and class-wildcard four-residue families are separate predefined diagnostics. Reference populations define enrichment; query tokens only look up those rules. The bounded motif factor is correlated sequence evidence, not an independent likelihood. V is counted once, and no J or length marginal likelihood is multiplied in. Neither S_g nor $S_{g,T}$ inherits the rank model's null bound or posterior. Biological call calibration and selected-label error remain separate tasks.

Remark 9 (prior-only categories)**.** If $n_g = m_g = 0$ the shared prior still yields $L_g \to (M+1)/(N+1)$ under unequal population sizes even though no germline rule was observed. That is a model-prior artefact, not evidence, and reference-defined activation must not credit it.

V. Parameters and why they take these values

Symbol / flag	Default	Role, admissible range and basis for the value
θ `--scope`	$(1, 0, 0, 1)$	Edit budget. Fixed, not fitted. Distance-1 neighbours are the measured signal:noise optimum (macro purity $0.49, 0.24, 0.15, 0.10, 0.07$ at edits 1..5). Any θ is admissible; cost grows sharply (Sec. 12).
λ `score_scale`	400	Exponential temperature on the VDJAM alignment score, in matrix units ($\times 100$). Inherited from the historical scorer, not re-fitted. Larger λ flattens the ranking toward a plain neighbour count.
matrix	VDJAM	Bundled TCR substitution table at integer scale 100; `none` gives unit edit cost. On held-out-epitope retrieval no amino-acid alphabet clearly beats BLOSUM62 and a genetic-code null ties it; *position* is the lever, not the alphabet.
w_V `soft_v`	on, 0.25	Same V family $\to$ 1; different family $\to$ 0.25$\times$ germline CDR1+CDR2 similarity; unknown $\to$ 1 (neutral). A V match is a measured $\sim$7$\times$ co-specificity prior, but continuous loop similarity does not recover it, so the cross-family weight is deliberately a small fixed discount rather than a fitted function.
+1 pseudocount	fixed	Control denominator $n_c + 1$: finite, size-invariant, and makes an exact hit counted rather than auto-accepted.
D `--radius`	90	Radius of the opt-in geometry route, in ordinary TCRdist units (native = 200$\times$). Determines computation, not a target-dependent gate. **Applied to $d_\alpha + d_\beta$ for linked pairs**, so the same numeral is a tighter per-chain constraint there (Remark 5).
bin width	200 native	Distance bins $\lceil d_{\text{native}}/200 \rceil$; both populations share the grid so ties are tied identically on both sides.
π	1/2	Prior for the rank-tilted posterior: equal *model* odds, explicitly not a biological prevalence.
δ	10^{-6}	Total failure budget across four simultaneous Clopper–Pearson category bounds in the punctured paired calibration; added to the reported tail.
gap cost	12/residue	Linear, not affine; 3$\times$ CDR3 and unit V-loop weights reproduce the TCRdist3 default geometry exactly.
`--threads`	1	Serial by default. The only parallel layer is the single native search batch; there is no worker pool around it.

Every entry above is a *fixed default*, not a data-derived or fitted quantity. The data-derived objects in the system are the bundled VDJAM table, the germline-retention profiles p_V, p_J (from the OLGA recombination model via `mirpy`) and the V-template loops — all versioned input artifacts, loaded, never re-fitted at run time.

VI. Performance bounds

Model and sizes. Unit-cost word RAM. Q distinct query junctions; N_{seq} distinct reference junctions and $N_{\text{rec}} \geq N_{\text{seq}}$ reference observation records for the locus; M control junctions; L the maximum junction length; $|\Sigma| = 20$; edit budget $t = t_{\max}$; R_D, R_C the numbers of returned reference and control hits (the *output* sizes).

Indexing. Trie construction over distinct junctions is $O((N_{\text{seq}} + M)L)$ time and space. It is paid once per reference/control, not per query.

Search. Bounded-edit trie traversal enumerates the t-neighbourhood of the query rather than scanning the reference. The number of distinct strings within edit distance t of a length-L string is $O\big((L|\Sigma|)^t\big)$, and each is resolved by an $O(L)$ walk, so per query the work is

$$O\big(L^{\,t+1}|\Sigma|^t\big) + O(\text{hits}), \qquad \text{independent of } N_{\text{seq}} \text{ and } M.$$

This is the central performance property: at the default $t = 1$ the search cost does not grow with the size of VDJdb. The whole-batch bound is $O\big(Q\,L^{t+1}|\Sigma|^t + R_D + R_C\big)$. The enumeration bound is worst-case and loose — the trie prunes — so the *measured* growth factor per extra edit is the operative number (Table 12).

Reduction. Hits expand from search keys to observation records with factor $\rho = N_{\text{rec}}/N_{\text{seq}}$ (measured: 1.50 TRB, 1.58 TRA on the 2026-06-11 human release), then group by (query, pMHC): $O(|H| \log |H|)$ with $|H| = \Theta(\rho R_D)$.

Space. $O((N_{\text{seq}} + M)L)$ for indices plus $O(|H|)$ for the hit table. The hit table, not the index, is the memory risk, and it is bounded by the output size, which is why a wide scope is a memory decision and not only a time decision.

Geometry route. `distance_matrix` is dense $\Theta(QN_{\text{seq}})$ int32 and is explicitly capped at 64 MiB per batch, so it batches at $\lfloor 64\,\text{MiB}/(4N_{\text{seq}}) \rfloor$ queries. The counting route (`total_distance_count_batch`) avoids the dense matrix: native work is $\Theta(QN_{\text{seq}}L)$ but output is only $O(Q \cdot |\text{cutoffs}|)$. The rank route is therefore *linear in reference size per query*, unlike the default search — this is the dominant cost asymmetry between the two routes and the reason the geometry arms need a resource gate.

Parallelism. One layer only: the single native search batch, threaded internally. No outer pool, no worker per query.

Measured, and whether it agrees

Human VDJdb 2026-06-11 (TRB $N_{\text{seq}} = 70{,}575$, $N_{\text{rec}} = 105{,}677$; TRA 43,227/68,206), control $M = 250{,}000$, 4,000 distinct TRB queries, one thread, Apple M-series:

Scope θ	target q/s	× cost	reference hits	control q/s	control hits
$1,0,0,1$	102,538	1.0	9,732	70,939	1,342
$2,0,0,2$	12,600	8.1	60,290	4,924	29,593
$2,1,1,2$	3,457	29.7	104,820	1,756	60,474
$3,0,0,3$	3,154	32.5	324,218	881	319,946

Index build 0.53 s for both loci; control load 0.02 s; full default evidence (both searches plus the whole polars reduction) 0.15 s for 4,000 queries, i.e. **26,724 queries/s single-threaded**, peak RSS 0.97 GB.

Agreement. The predicted independence from reference size holds: 70,575 reference junctions are searched at 10^5 q/s. The predicted sharp growth in t holds in direction, and the measured factors (8.1× then 4.0× for the target; 14.4× then 5.6× for the control) are far below the worst-case enumeration factor $L|\Sigma| \approx 600$ — the trie prunes by roughly two orders of magnitude, so the asymptotic bound is correct but should not be used for capacity planning. The reduction is ≈ 37% of default wall time, so neither search nor reduction is negligible and the practical bottleneck shifts to the reduction only as ρ or the candidate count grows. Thread scaling of the single native batch at scope $2,1,1,2$ over 20,000 queries: $1 \to 5.62$ s, $2 \to 2.89$ s (1.95×), $4 \to 1.44$ s (3.90×), $8 \to 0.75$ s (7.53×) — efficiency 0.94–0.98, so the declared single parallel layer is real and not overhead. Recorded full-benchmark costs are consistent in order of magnitude with the geometry route's linear-in-N_{seq} behaviour: the curated all-task rebuild took 776.6 s/4.47 GB (original/native), 407.2 s/3.21 GB (prior rank) and 977.9 s/7.94 GB (current arm) over 615,224 case appearances.

VII. Accuracy: what can be bounded, and what is observed

What is exactly guaranteed. Exactly one accuracy statement here is a theorem: Proposition 1 gives $\Pr_0(p_{\text{bound}} \le \alpha) \le \alpha$, a valid but conservative type-I bound for the supported-query statistic under exchangeable labels. Nothing else below is a guarantee.

What is conditional. The Poisson enrichment tail is valid to Chen–Stein error under the background model; the punctured paired bound is valid up to its declared $\delta = 10^{-6}$ under IID category sampling and independent background chains. Both assumptions are stated, neither is verified by the data: deduplicated controls and generated-draw multiplicity do not establish exchangeability, and this is the appendix's principal unverified premise.

Why no useful bound on top-1 accuracy exists. A distribution-free bound would require a margin assumption linking distance to co-specificity. The data refuse it: co-specificity is not a monotone function of distance with a usable gap. The honest substitute is the measured purity curve — the fraction of neighbours at a given distance that share an epitope — which is the Bayes ceiling of any rule that uses distance alone. Macro purity (per-epitope mean, human TRB) is $0.49, 0.24, 0.15, 0.10, 0.07$ at edit distance 1..5, an enrichment of $56\times \to 2.5\times$ over chance.

Any scorer confined to a radius-t ball inherits that ceiling; this is why the search radius, not the substitution matrix, is the first-order lever, and why the matrix can only re-rank within a ball whose purity the radius has already fixed.

A derived consequence, and its measurement. The first-hit calibration rests on the claim that the background ball fills faster than the target ball. That is checkable. At $N/M = 0.282$ the measured enrichment n_D/E falls from 25.7× at $t = 1$ (9,732 reference vs 1,342 control hits) to 3.6× at $t = 3$ (324,218 vs 319,946): the control count grows 238× while the reference count grows 33×. The mechanism the adaptive scope relies on is therefore real and quantified.

The degeneracy, measured. Proposition 2 predicts that queries with $n_D = 0$ are ranked by background crowding alone, so accuracy should degrade with target support. It does, monotonically, and for *every* geometry arm including the unmodified native TCRdist3 comparator. Mean AP difference against the shipped scorer over 620 curated tasks, by the number of positives in the task:

Arm	$[30, 100)$ (328 tasks)	$[100, 300)$ (145)	$[300, 1000)$ (95)	≥ 1000 (48)
distance-local rank (conditional)	−0.0636	−0.0244	+0.0065	+0.0104
uncensored nearest rank	−0.0673	−0.0302	+0.0004	+0.0253
native TCRdist3 default	−0.0749	−0.0399	−0.0054	−0.0007

Case-level confirmation at radius $D = 90$, $M \approx 1.1 \times 10^5$ (human TRB): for ALWALPHAA ($N = 30$) the median query has one reference inside D and 12/31 positives and 15/31 negatives have none; the Bayes-factor medians are 0.4112 (positives) and 0.4174 (negatives) — statistically indistinguishable, AP 0.8841 → 0.4893, i.e. chance for a balanced task. For CINGVCWTV ($N = 386$) the medians are 1.4397 vs 0.5612; for GILGFVFTL ($N = 6{,}686$), 2.8178 vs 0.7514. Restricting each task to its supported subpopulation ($n_D \geq 1$, the support boundary of the statistic, not a fitted threshold) removes about half the deficit at low support, flips the sign at ≥ 300 positives, and removes the *entire* linked-pair deficit (−0.1408 → +0.0005 over 86 paired tasks). Prediction and observation agree; the agreement is not itself proof of the mechanism, but no competing explanation accounts for the monotonicity, the sign flip and the paired-specific magnitude simultaneously.

Where the linked-pair metric is actually decided. On the eight priority targets, when a linked reference pair lies inside the radius both the shipped scorer and the rank arm reach AP 0.9984–1.0000 on eleven of fourteen tasks. The metric is therefore decided on the 52–92% of queries that have no supported linked reference. For linked-pair annotation of well-populated targets the binding constraint is **coverage, not the score**. Support is also an unusually pure indicator in itself: over the fourteen priority linked-pair tasks at $D = 90$ a linked reference pair lies inside the radius for 0.593 of positives against 0.057 of negatives, a 10.4× purity ratio — which is why ordering by support first (Remark 7) recovers so much on linked pairs and is nearly inert on single chains, where the median supported fraction is 0.993. Widening the radius trades that purity for reach monotonically: 0.725 against 0.173 (4.2×) at $D = 120$, 0.850 against 0.425 (2.0×) at $D = 150$, and by $D = 240$ both populations are fully covered and the indicator carries nothing. The separation is broad and nearly flat from $D = 90$ to $D = 150$ while the ratio falls steeply, so this is a genuine trade-off, not an optimum to be read off. (The supported subpopulation is easier by construction, so these are not deployable AP values; the decomposition — which population decides the metric — is the result.)

Remark 10 (acceptance geometry)**.** Because the per-task delta is monotone in support and the eight priority targets all lie above the crossover, a pooled all-task mean and a priority-target mean can disagree in *sign* for the same arm. They do: over the 70 priority tasks the conditional-rank arm gains +0.0213 mean AP (+0.0697 on the 14 linked-pair tasks, worst-case AP 0.5082 → 0.7298), while over all 620 tasks it loses −0.0391. Reporting only the pooled mean therefore selects against the stated objective. Both must be reported.

VIII. Open obligations

1. **Exchangeability is assumed, not shown.** Reference deduplication, generated-draw multiplicity and study ascertainment all threaten it. Every p-value and posterior here is conditional on it.
2. **The nearest-conditioned branch has no null.** $\mathbb{E}_0 B_X^{\text{cond}} = 1$ is unproved (Remark 8); p_{bound} must not be claimed for queries routed through it.
3. **The $n_D = 0$ branch is a design question.** Proposition 2 shows the imputation is uninformative by construction. Reporting B_X conditional on supported references and carrying n_D as explicit evidence

forfeits Proposition 1 *and* is measurably worse (Remark 7); the informative change is to the disposition of zero-support queries, which leaves B_X and its null untouched. Making it is the author's call.

4. **The paired radius and the single-chain radius are not the same quantity** and currently share a numeral (Remark 5).
5. **No biological calibration.** p_{bound} is not Pr(FP) and the posterior is not Pr(TP). The empirical any-call operating point is a separate deployment artifact.
6. **The low-support regime is unexplained.** Below roughly 100 positives every geometry arm tested, including the unmodified native TCRdist3 comparator, stays behind the original density scorer, and three rounds of reformulating the rank statistic did not move it. That it is common to every geometry suggests a property of the regime rather than of any one construction, but no mechanism has been established and none should be asserted.
7. **Occam.** Twelve scoring routes are maintained in parallel and every candidate has been rejected for promotion; the default is unchanged. Each should be promoted or retired behind its recorded run artifacts.

Original scoring remains default; M3 stays deferred.